

# THE MATHEMATICS OF MAYBE

Grade 9 — Introduction to Probability | Answer Key  
Made under ChauhanBooks

**Note:** Questions that require an actual experiment (such as tossing a coin or a paper cup) can have different experimental answers. For those, a valid sample result/formula is given.

## Exercise Set 7.1

- 1(i).** Certain — probability = 1. Monday always comes after Sunday.
- 1(ii).** Impossible / extremely unlikely in the usual school-level interpretation — probability = 0. Mumbai does not normally receive snow in July.
- 1(iii).** Impossible — probability = 0. An elephant will not ordinarily walk through the classroom today.
- 1(iv).** More likely — the exact probability depends on the circumstances, but greeting at least one friend at school tomorrow is more likely than not for a typical school day.

## Exercise Set 7.2

- 1(i).** Green probability =  $8/30 = 4/15 \approx 0.267$  (26.7%).
- 1(ii).** Yellow proportion =  $7/30$ . Estimated yellow sweets =  $600 \times 7/30 = 140$ .
- 2(i).** Arts Club probability =  $11/40 = 0.275$  (27.5%).
- 2(ii).** Sports Club proportion =  $9/40$ . Estimated students =  $800 \times 9/40 = 180$ .
- 3(i–iii).** These depend on the 20 actual tosses. If heads occur  $h$  times, tails occur  $20 - h$  times, and experimental  $P(H) = h/20$ .
- 3(iv).** For a fair coin,  $P(T) = 1/2 = 0.5$  (50%).
- 4.** Record the numbers of bottom, top and side landings as  $b$ ,  $t$  and  $s$ , where  $b+t+s=100$ . Then  $P(\text{bottom})=b/100$ ,  $P(\text{top})=t/100$ ,  $P(\text{side})=s/100$ .
- 5.** Even outcomes are  $\{2,4,6\}$ .  $P(\text{even}) = 3/6 = 1/2$ .
- 6(i).** Experimental probability =  $3/12 = 1/4 = 0.25$ .
- 6(ii).** Theoretical probability =  $1/6 \approx 0.167$ .
- 6(iii).** They may differ because experimental results vary by chance. As the number of trials becomes large (60, 600, 6000), experimental probability is expected to get closer to the theoretical probability (Law of Large Numbers).

## Exercise Set 7.3

- 1.** Sample size = 6.
- 2(i).**  $S = \{(1,H),(1,T),(2,H),(2,T),(3,H),(3,T),(4,H),(4,T),(5,H),(5,T)\}$ .
- 2(ii).**  $S = \{-5,-4,-3,-2,-1,0,1,2,3,4,5\}$ .
- 2(iii).** If outcomes are recorded by colour:  $S = \{\text{Green, Red}\}$ .
- 3(i).**  $S = \{(\text{Samosa,Chai}),(\text{Samosa,Lassi}),(\text{Pakora,Chai}),(\text{Pakora,Lassi}),(\text{Bhaji,Chai}),(\text{Bhaji,Lassi})\}$ .
- 3(ii).** Event =  $\{(\text{Samosa,Chai}),(\text{Samosa,Lassi})\}$ .

## Exercise Set 7.4

- 1(i).** Tree branches: Basket A  $\rightarrow$  Apple ( $1/3$ ), Orange ( $2/3$ ); Basket B  $\rightarrow$  Banana ( $1/2$ ), Mango ( $1/2$ ).

**1(ii).**  $S = \{(Apple, Banana), (Apple, Mango), (Orange, Banana), (Orange, Mango)\}$ .

**1(iii).**  $P(\text{Apple and Banana}) = (1/3)(1/2) = \mathbf{1/6}$ .

**2(i).** Colour outcomes:  $\{RR, RB, BR, BB, RG, GR, BG, GB, GG\}$ . Since the pen is replaced, each draw has probabilities  $R=3/9$ ,  $B=4/9$ ,  $G=2/9$ .

**2(ii).** Same colour:  $RR, BB, GG$ .  $P = (3/9)^2 + (4/9)^2 + (2/9)^2 = 9/81 + 16/81 + 4/81 = \mathbf{29/81} \approx \mathbf{0.358}$ .

## End-of-Chapter Exercises

1(i). 0.

1(ii). Sample space.

1(iii). 1.

1(iv).  $\frac{1}{2}$ .

2. Frequency = 15; relative frequency =  $15/50 = \frac{3}{10} = 0.3$ .

3(i). Not necessarily. Starting a car is not guaranteed to have equally likely outcomes.

3(ii). Yes, for a fair coin: H and T are equally likely.

3(iii). Yes, for a fair die: 1–6 are equally likely.

3(iv). No, because red and blue are not equally likely ( $3/10$  vs  $7/10$ ).

3(v). The textbook wording treats boy/girl as the two outcomes, but equal likelihood is an assumption rather than something guaranteed by the statement alone.

4(i).  $S = \{HH, HT, TH, TT\}$ ; at least one head =  $\{HH, HT, TH\}$ .  $P = \frac{3}{4}$ .

4(ii).  $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ ; even =  $\{2, 4, 6, 8, 10\}$ .  $P = \frac{5}{10} = \frac{1}{2}$ .

4(iii).  $S = \{1, 2, 3, 4, 5, 6\}$ ; greater than 4 =  $\{5, 6\}$ .  $P = \frac{2}{6} = \frac{1}{3}$ .

4(iv). Total balls = 6; not red =  $\{\text{blue, blue, green}\} = 3$  balls.  $P = \frac{3}{6} = \frac{1}{2}$ .

4(v). Exactly two heads:  $\{HHT, HTH, THH\}$ .  $P = \frac{3}{8}$ .

5.  $S = \{\text{strawberry, lemon, mint}\}$ ;  $P(\text{strawberry}) = \frac{1}{3}$ .

6. Combinations: Red–Jeans, Red–Khakis, Red–Shorts, Blue–Jeans, Blue–Khakis, Blue–Shorts. Total = 6.

7(i).  $20/1000 = \frac{1}{50} = 0.02$ .

7(ii). 4000–14000 km:  $(210+325)/1000 = \frac{535}{1000} = \frac{107}{200} = 0.535$ .

7(iii).  $445/1000 = 0.445 = 44.5\%$ .

8(i). PEACE has 5 cards. P, E or C occurs on 3 cards (P, one of the E cards? No: the letters are P, E, A, C, E, so distinct requested cards are P, E, C = 4 cards because E appears twice). Thus  $P = \frac{4}{5}$ .

8(ii). Not E = P, A, C = 3 cards.  $P = \frac{3}{5}$ .

9(i).  $P(8) = \frac{1}{8}$ .

9(ii). Odd numbers  $\{1, 3, 5, 7\}$ :  $P = \frac{4}{8} = \frac{1}{2}$ .

9(iii). Greater than 2:  $\{3, 4, 5, 6, 7, 8\}$ :  $P = \frac{6}{8} = \frac{3}{4}$ .

9(iv). Every outcome is less than 9:  $P = 1$ .

9(v). Multiples of 3:  $\{3, 6\}$ :  $P = \frac{2}{8} = \frac{1}{4}$ .

10(i). Red then blue =  $(\frac{4}{9})(\frac{5}{8}) = \frac{5}{18}$ .

10(ii). Two blue =  $(\frac{5}{9})(\frac{4}{8}) = \frac{5}{18}$ .

11. An event with probability 0: sum of two dice is 13. An outcome/event with probability 1: sum is **between 2 and 12**.

12(i). Prime sums greater than 5 are 7 and 11. There are 6 ways for 7 and 2 ways for 11, so  $P = (6+2)/36 = \frac{2}{9}$ .

12(ii). Different colours = 1 – same colour. Same =  $(\frac{4}{9})(\frac{3}{8}) + (\frac{3}{9})(\frac{6}{8}) + (\frac{2}{9})(\frac{7}{8}) = \frac{1}{6} + \frac{1}{4} + \frac{7}{36} = \frac{5}{9}$ . Hence  $P(\text{different}) = \frac{4}{9}$ .

**12(iii).** First coin H and exactly two heads total means outcomes HHT, HTH.  $P = \frac{2}{8} = \frac{1}{4}$ .

**12(iv).** Even numbers end in 2 or 4. Total  $4! = 24$ ; favourable  $= 2 \times 3! = 12$ .  $P = \frac{1}{2}$ .

**12(v).** Exactly 2 correct out of 3 guesses:  $C(3,2)(\frac{1}{4})^2(\frac{3}{4}) = \frac{9}{64}$ .

**13(i).** With replacement:  $S = \{(1,1), (1,2), \dots, (4,4)\}$ ; **16** outcomes.

**13(ii).** Without replacement:  $S = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$ ; **12** outcomes.

**13(iii).** Sample-space sizes: **16** and **12**.

**14.**  $S = \{(H,1), (H,2), (H,3), (H,4), (H,5), (H,6), (T,1), (T,2), (T,3), (T,4), (T,5), (T,6)\}$ ; size=**12**.

**15.** (iv) **{0,1,2,3}** is the correct sample space. (i) misses 0; (ii) misses 3; (iii) contains impossible outcome 4.

**16.** Rectangle area  $= 2 \times 3 = 6 \text{ m}^2$ . Circle diameter  $= 1 \text{ m}$ , so radius  $= 0.5 \text{ m}$  and area  $= \pi(0.5)^2 = \pi/4$ . Probability  $= (\pi/4)/6 = \pi/24 \approx \mathbf{0.131}$  (13.1%).

## Quick Formula Sheet

Experimental probability = number of times the event occurred  $\div$  total number of trials.

Theoretical probability = number of favourable outcomes  $\div$  number of possible outcomes.

For a valid probability:  $0 \leq P(E) \leq 1$ .